Advanced Computer-Aided VLSI System Design
Lab7: Physical Verification

TA: 蔡岳峰 f12943014@ntu.edu.tw

Data Preparation

1. The extracted directory contains
   - **Library (technology-dependent files)**
     - **iPDK**
     - **encrypted design rules**
   - **design**
     A. CHIP.gds
     B. CHIP_pg.v
   - **ipMerge:** the workspace for merging GDSs of IPs into the chip GDS
   - **insertBeDummy:** the workspace for creating back-end-of-line dummy
   - **insertFeDummy:** the workspace for creating front-end-of-line dummy
   - **dummyMerge:** the workspace of the DRC check
   - **drc:** the workspace of the DRC check
   - **v2lvs:** the workspace for converting the Verilog file to a SPICE file
   - **lvs:** the workspace of the LVS check
   - **virtuoso:** the workspace for checking the layout of the GDS files

Introduction

In this lab, you will learn how to use the PDK generator from TSMC, use Calibre to merge the IP from the IP provider, generate the dummy metal for a chip design, and run the DRC and LVS checks in Calibre in ADFP technology.

Environment Setting

1. Source the CIC C shell files on the ADFP workstation
   - `fc`
2. Set up the Cadence initialization file **".cdsinit"** to automatically include Calibre in the Virtuoso for further sign-off verification
   - Create a file ".**cdsinit**" in the home directory.
   - Add the contest below

```
setSkillPath(". ~ /share1/cad/mentor/calibre/cur/shared/pkgs/icv/tools/queryskl")
load("calibre.OA.skl")
```

3. Copy the lab7 file to your directory

- `cp /share1/home/acvsd-ta11/lab7.tar.gz .`
- `tar -zxvf lab7.tar.gz`
- `cd lab7`

Generate PDK

1. Open a terminal and change to the Virtuoso directory under the lab folder
   - `cd virtuoso`
2. Copy N16ADFP_iPDK
   - `cp -r /share1/tech/ADFP/Executable_Package/Collaterals/Tech/iPDK/N16ADFP_iPDK/ .`
3. Change to N16ADFP_iPDK directory
   - `cd N16ADFP_iPDK`
4. Invoke the iPDK installation script
   - `./pdkInstall.pl`
5. Configure the iPDK setting
   - Available PDK types: 2 – Skill Pcell
   - Available process types: 1 – L0
   - Available voltages: 1 – 0.8/1.8V
   - Available metal options: 1 – 1P11M_2Xa1Xd_h_3Xe_vhv_2Y2R

```
*Available PDK types: (1,2)
************** Please choose PDK type  ***********************
   1 - PyCell            : support only Synopsys platform
   2 - Skill Pcell       : support only Cadence platform
============================================================
Please enter your choice: (1/2) :
2
*Available process types are:
   1 - L0
Please enter your choice: (1,2...)
1
*Available voltages are:
   1 -  0.8/1.8V   [L0]
Please enter your choice: (1,2...)
1
*Available metal options are:
   1 - 1P11M_2Xa1Xd_h_3Xe_vhv_2Y2R UT-AlRDL
   2 - 1P11M_2Xa1Xd_v_3Xe_hvh_2Y2R UT-AlRDL

Please enter your choice: (1,2...)
1
```

6. Confirm with your selection **(Checkpoint 1)**

```
Please confirm with your selction information:
*********************************************************
*** Select PDK type : Skill Pcell
*** Select process : L0
*** Select voltage : 0.8/1.8V
*** Select metal option : 1P11M_2Xa1Xd_h_3Xe_vhv_2Y2R UT-AlRDL
*********************************************************
Are these correct (y|n) ?
y
```

7. Run Virtuoso (no need to close the virtuoso in this lab)
   - `virtuoso &`

Merge IP

1. Open a terminal and change to the ipMerge directory under
   - `cd ipMerge`
2. Check if the path to the block GDS in scr/var.tcl is correct
3. Add the path to the new IP to the IP in scr/var.tcl list if needed

```
set blockGds ../design/CHIP.gds
set gdsList  " \
    $ProjRoot/Executable_Package/Collaterals/IP/stdcell/N16ADFP_StdCell/GDS/N16ADFP_StdCell.gds \
    $ProjRoot/Executable_Package/Collaterals/IP/sram/N16ADFP_SRAM/GDS/N16ADFP_SRAM_100a.gds \
    $ProjRoot/Executable_Package/Collaterals/IP/pll/N16ADFP_PLL/GDS/n16adfp_pll_100a.gds \
    $ProjRoot/Executable_Package/Collaterals/IP/stdio/N16ADFP_StdIO/GDS/N16ADFP_StdIO.gds \
    $ProjRoot/Executable_Package/Collaterals/IP/bondpad/N16ADFP_BondPad/GDS/N16ADFP_BondPad.gds \
"
```

4. Read the script scr/runset.cmd

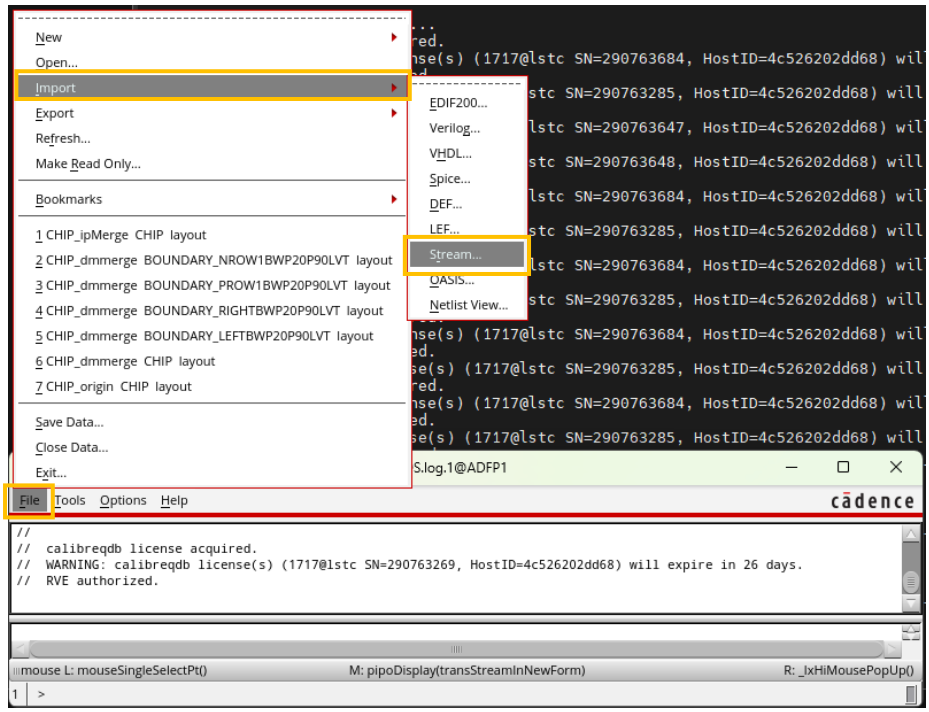
```
# Read the top-level block gds
set top [layout create $blockGds -dt_expand -preservePaths -preserveTextAttributes \
    -preserveProperties]

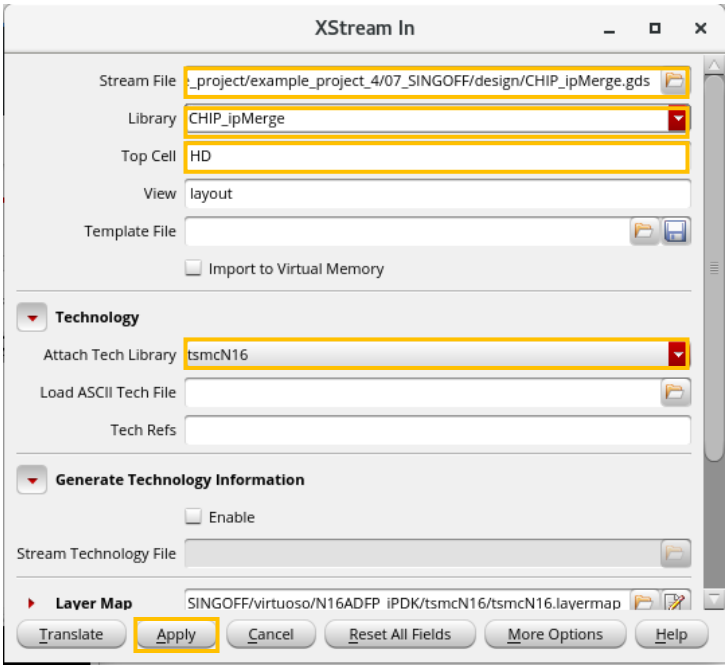
# Recursively load and merge the GDS of IPs
foreach gdsFile $gdsList {
    # Load IP GDS
    set toImport [layout create "$gdsFile" -dt_expand -preservePaths \
        -preserveTextAttributes -preserveProperties]

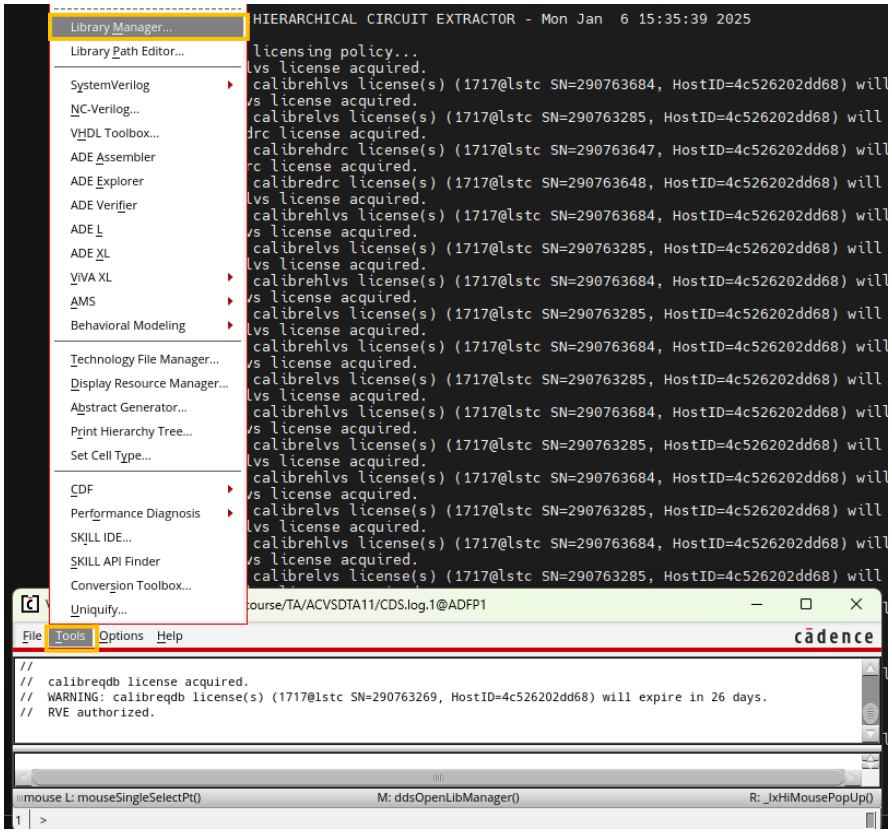
    # Merge IP GDS
    $top import layout $toImport FALSE overwrite -dt_expand -preservePaths \
        -preserveTextAttributes -preserveProperties
}

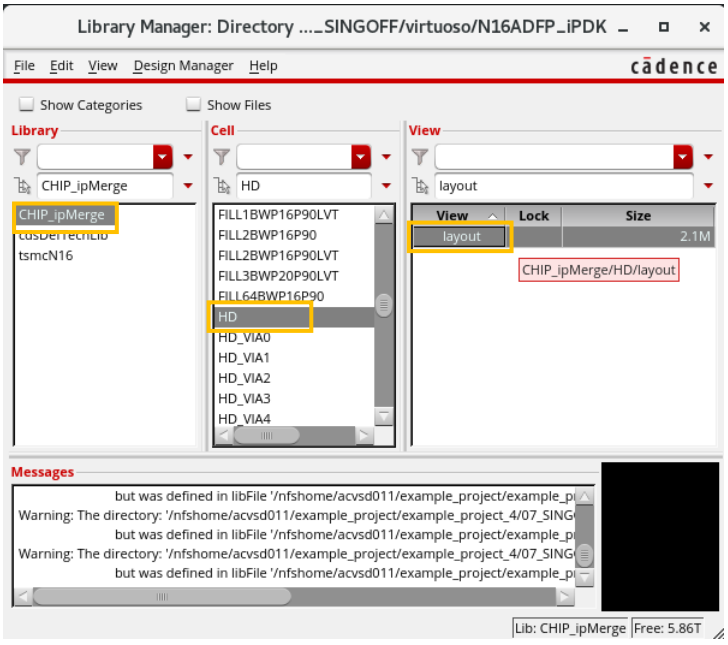
# Write GDS File
$top gdsout ./output/CHIP_ipMerge.gds $TopCell
$top gdsout ../design/CHIP_ipMerge.gds $TopCell
```

5. Run CalibreDRV to merge IP
   - `./run.csh`
6. In virtuoso, stream in the GDS file
   - Stream in the GDS file and create the library called CHIP_ipMerge
   - The attach tech library should select tsmcN16
   - Open the layout from the library manager (**Checkpoint 2**)

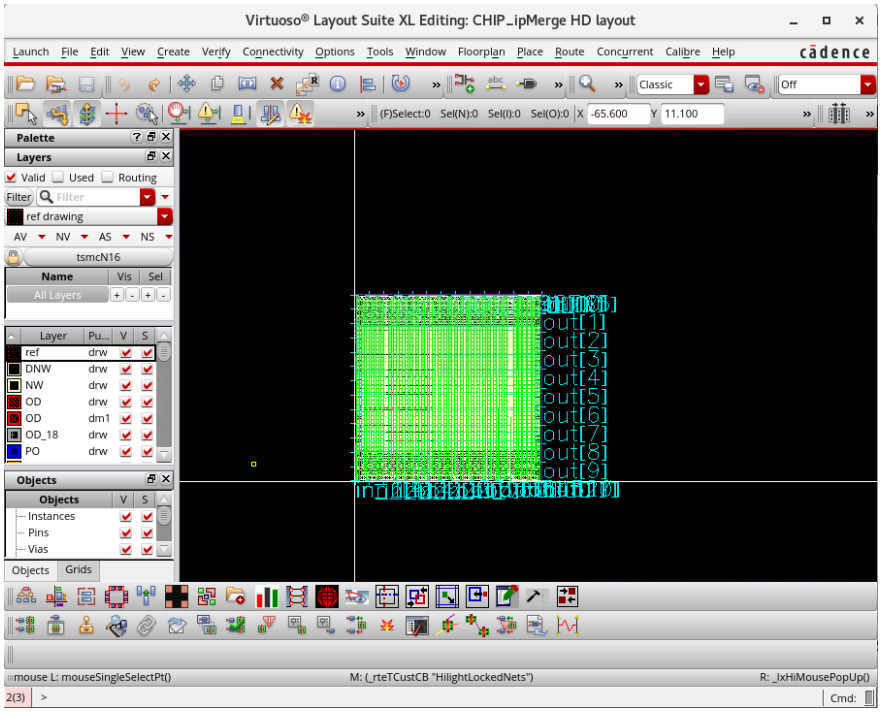

New
Open...
Import
Export
Refresh...
Make Read Only...
Bookmarks
1 CHIP_ipMerge CHIP layout
2 CHIP_dmmerge BOUNDARY_NROW1BWP20P90LVT layout
3 CHIP_dmmerge BOUNDARY_PROW1BWP20P90LVT layout
4 CHIP_dmmerge BOUNDARY_RIGHTBWP20P90LVT layout
5 CHIP_dmmerge BOUNDARY_LEFTBWP20P90LVT layout
6 CHIP_dmmerge CHIP layout
7 CHIP_origin CHIP layout
Save Data...
Close Data...
Exit...
EDIF200...
Verilog...
VHDL...
Spice...
DEF...
LEF...
Stream...
OASIS...
Netlist View...
File Tools Options Help
cadence
// calibreqdb license acquired.
// WARNING: calibreqdb license(s) (1717@lstc SN=290763269, HostID=4c526202dd68) will expire in 26 days.
// RVE authorized.
mouse L: mouseSingleSelectPt()
M: pipoDisplay(transStreamInNewForm)
R: _lxHiMousePopUp()


XStream In
Stream File _project/example_project_4/07_SINGOFF/design/CHIP_ipMerge.gds
Library CHIP_ipMerge
Top Cell HD
View layout
Template File
Import to Virtual Memory
Technology
Attach Tech Library tsmcN16
Load ASCII Tech File
Tech Refs
Generate Technology Information
Enable
Stream Technology File
Layer Map SINGOFF/virtuoso/N16ADFP_iPDK/tsmcN16/tsmcN16.layermap
Translate
Apply
Cancel
Reset All Fields
More Options
Help





7. Check if the IPs are merged from the library manager



Generate Front End Dummy

1. Change to insertFeDummy directory
   - ➢ **`cd insertFeDummy`**
2. Read the script scr/run.csh

```
# Copy the encrypted FEOL dummy rule deck to the local scr directory
cp -rf \
    "$ProjRoot/…/N16ADFP_Dummy_Calibre/FEOL/Dummy_FEOL_CalibreYE_16nm_ADFP_FFP.10a.encrypt" \
    ./scr/Dummy_FEOL_CalibreYE_16nm_ADFP_FFP.10a_eval122121.encrypt.modified

# Modify the copied FEOL dummy rule deck (e.g. LAYOUT PATH… -> //LAYOUT PATH…)
sed -i -e 's/^LAYOUT PRIMARY/\/\/LAYOUT PRIMARY/g' ./scr/Dummy_FEOL….encrypt.modified
sed -i -e 's/^LAYOUT PATH/\/\/LAYOUT PATH/g' ./scr/Dummy_FEOL….encrypt.modified
sed -i -e 's/^LAYOUT SYSTEM/\/\/LAYOUT SYSTEM/g' ./scr/Dummy_FEOL….encrypt.modified
sed -i -e 's/^DRC RESULTS DATABASE/\/\/DRC RESULTS DATABASE/g' ./scr/Dummy_….encrypt.modified
sed -i -e 's/^DRC SUMMARY REPORT/\/\/DRC SUMMARY REPORT/g' ./scr/Dummy_FEOL….encrypt.modified

# Disable the sealring option and enable custom design boundary
sed -i -e 's/ #DEFINE WITH_SEALRING/\/\/#DEFINE WITH_SEALRING/g' ./scr/….encrypt.modified
sed -i -e 's/\/\/#DEFINE UseprBoundary/#DEFINE UseprBoundary/g' ./scr/Dummy….encrypt.modified
```

3. Read the script scr/runset.cmd

```
# Specify the type of the layout, the path to the layout, and the top module of the layout
LAYOUT SYSTEM GDS
LAYOUT PATH "../design/CHIP_ipMerge.gds"
LAYOUT PRIMARY "HD"

# Specify output files
DRC RESULTS DATABASE "output/FEOL.db"
DRC SUMMARY REPORT "output/FEOL.sum"

# Read Modified Encrypted FEOL Rule
include ./scr/Dummy_FEOL_CalibreYE_16nm_ADFP_FFP.10a_eval122121.encrypt.modified
```

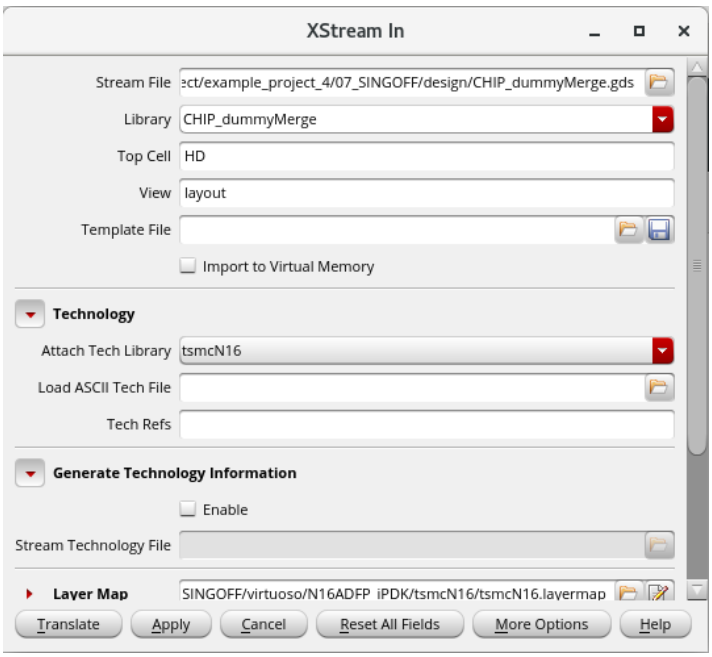
4. Run Calibre to generate dummy filler
   - `./run.csh`

Generate Back End Dummy

1. Change to insertBeDummy directory
   - `cd insertBeDummy`
2. Run Calibre to generate dummy metal
   - `./run.csh`

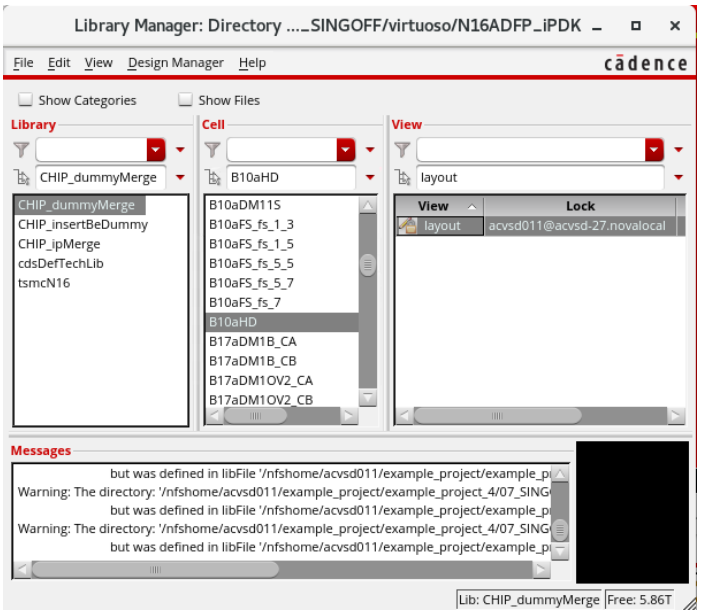
Merge Dummy

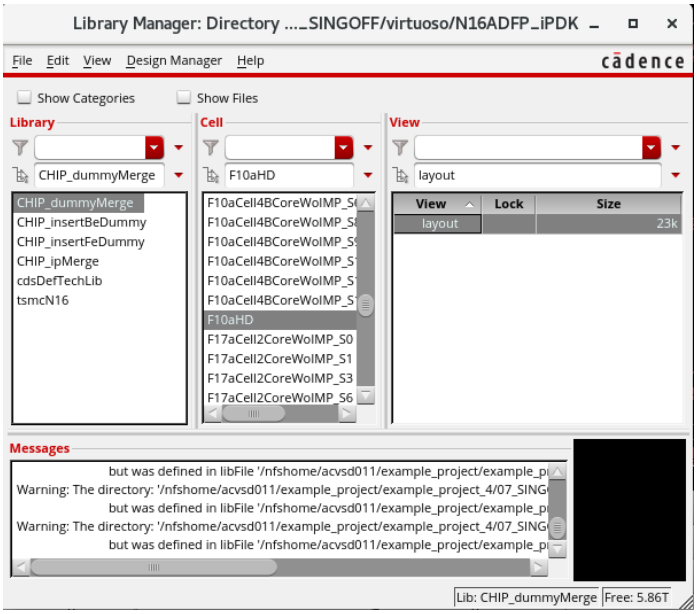
1. Change the directory to dummyMerge directory
   - `cd dummyMerge`
2. Run Calibre to merge GDS files
   - `./run.csh`
3. In Virtuoso, stream in the GDS file
   - Stream in the GDS file and create the library called CHIP_dummyMerge



4. Check if the dummies are merged from the library manager (Checkpoint 3)

➢ Room into the bottom-left corner to check if the dummy is merged correctly



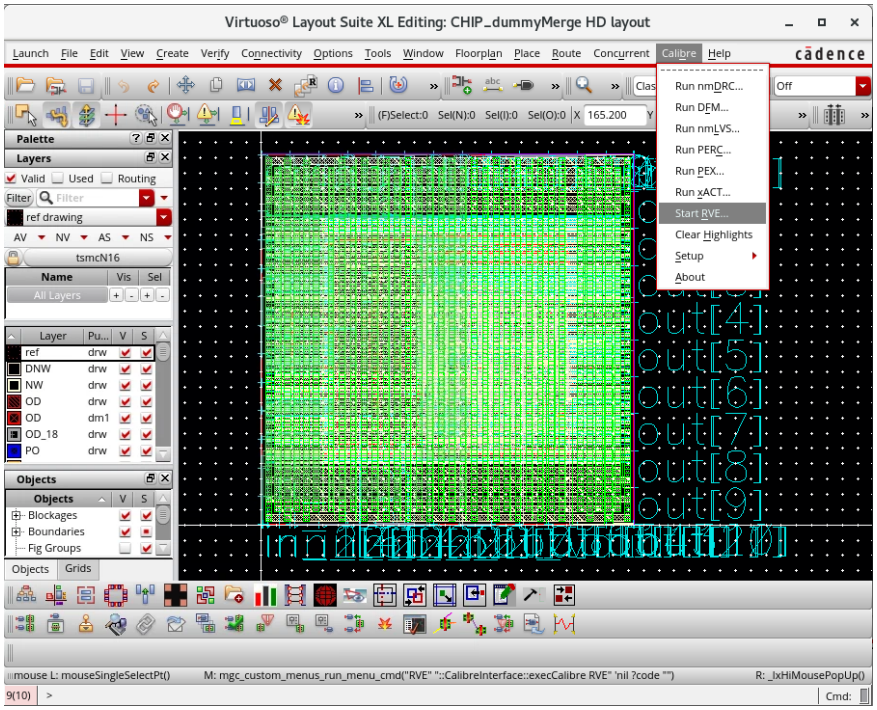


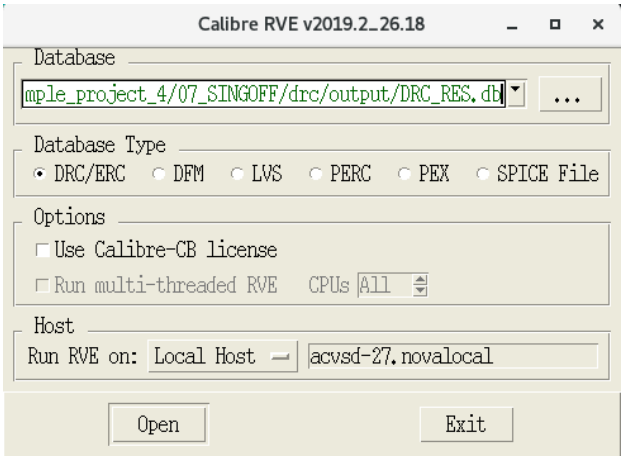
Design Rule Check

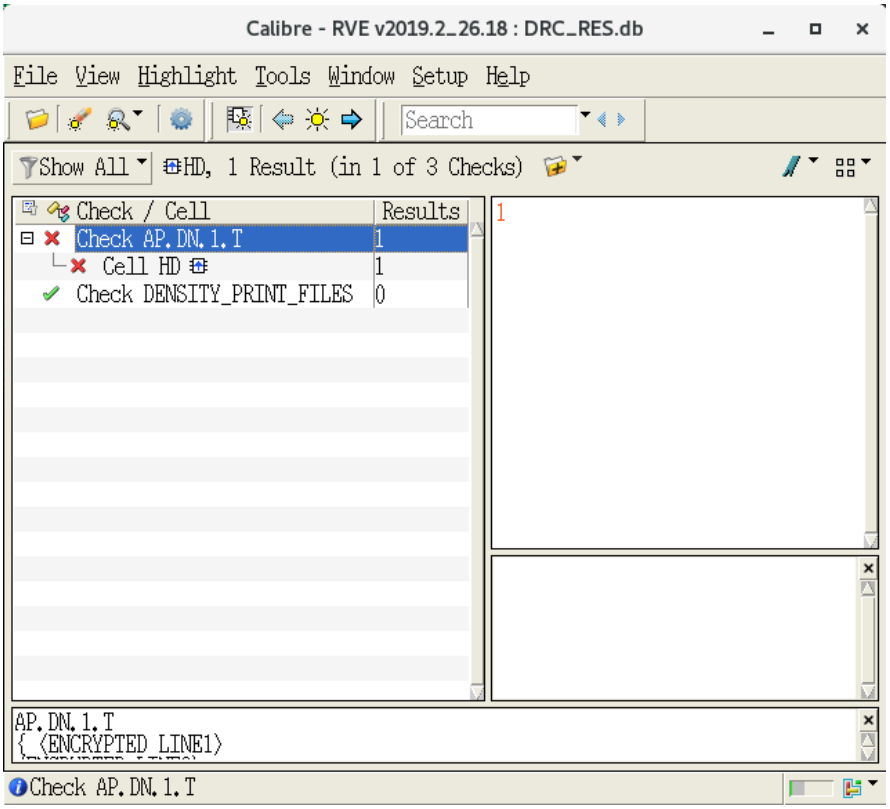
1. Change the directory to the drc directory
   - `cd drc`
2. Please check the script scr/runset.cmd

```
# Specify the type of the layout, the path to the layout, and the top module of the layout
LAYOUT SYSTEM GDS
LAYOUT PATH "../design/CHIP_dummyMerge.gds"
LAYOUT PRIMARY "HD"
```

3. Run the DRC check
   - `./run.csh`
4. Check the DRC result (**Checkpoint 4**)
   - Open the dummyMerge layout in Virtuoso
   - Start the Calibre RVE browser from Virtuoso
   - Load the database document into the RVE
   - Check and fix the DRC (if needed)


Virtuoso® Layout Suite XL Editing: CHIP_dummyMerge HD layout
Launch File Edit View Create Verify Connectivity Options Tools Window Floorplan Place Route Concurrent Calibre Help
cadence
Run nmDRC...
Run DFM...
Run nmLVS...
Run PERC...
Run PEX...
Run xACT...
Start RVE...
Clear Highlights
Setup
About
Palette
Layers
Valid Used Routing
ref drawing
tsmcN16
Objects
Blockages
Boundaries
Fig Groups
out[4]
out[5]
out[6]
out[7]
out[8]
out[9]


Calibre RVE v2019.2_26.18
Database
mple_project_4/07_SINGOFF/drc/output/DRC_RES.db
...
Database Type
DRC/ERC DFM LVS PERC PEX SPICE File
Options
Use Calibre-CB license
Run multi-threaded RVE CPUs All
Host
Run RVE on: Local Host
acvsd-27.novalocal
Open
Exit



Layout Versus Schematic Check

1. Change the directory to v2lvs directory
   - ➢ **cd v2lvs**
2. Add the IP list to the ./scr/var.tcl (optional)

```
set ProjRoot = "/share1/tech/ADFP"
set spilist = "\
  $ProjRoot…/IP/stdcell/N16ADFP_StdCell/SPICE/N16ADFP_StdCell_100b.spi \
  $ProjRoot…/IP/sram/N16ADFP_SRAM/SPICE/N16ADFP_SRAM_100a.spi \
  $ProjRoot…/IP/pll/N16ADFP_PLL/SPICE/n16adfp_pll_100a.spi \
  $ProjRoot…/IP/stdio/N16ADFP_StdIO/SPICE/N16ADFP_StdIO.spi \
  $ProjRoot…/IP/sram/N16ADFP_SRAM/SPICE/N16ADFP_SRAM_100a.spi \
"
```

3. Run v2lvs
   - ➢ **/run.csh**
4. Change the directory to lvs directory
   - ➢ **cd lvs**
5. Add the IP block to the ./scr/runset.cmd (optional)

```
395  LVS BOX PVDD1CDGM_V  PVDD1CDGM_V
1396  LVS BOX PVDD2CDGM_H  PVDD2CDGM_H
1397  LVS BOX PVDD2CDGM_V  PVDD2CDGM_V
1398  LVS BOX N16ADFP_PLL  N16ADFP_PLL

1400  include ./scr/N16ADFP_LVS_Calibre.modified
```

6. Modify the gds path in scr/runset.cmd as follows

```
VARIABLE POWER_NAME  "VDD"
VARIABLE GROUND_NAME "VSS"

LAYOUT PRIMARY "HD"
# The GDS should include the GDS of IPs
LAYOUT PATH "../design/CHIP_dummyMerge.gds"
LAYOUT SYSTEM GDS
```

7. Run ./run.csh to run layout versus schematic
   - `./run.csh`
8. Check that the LVS result (output/lvs.rep) is passed (**Checkpoint 5**)

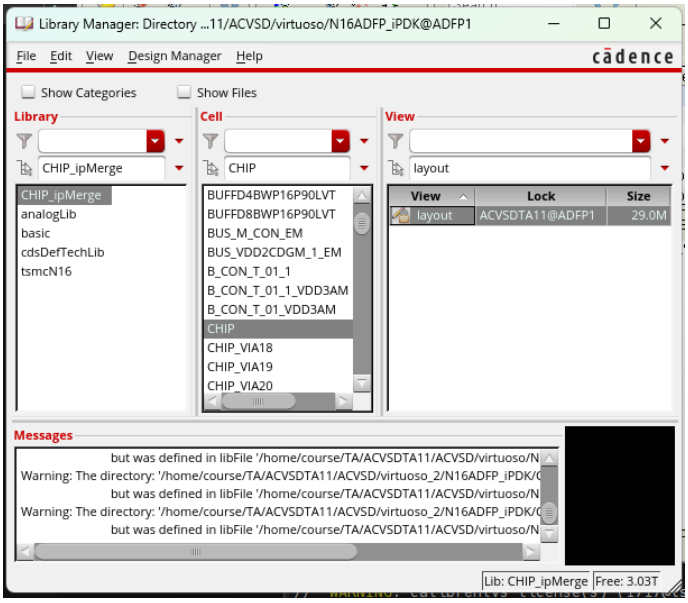
```
259
260                    CELL COMPARISON RESULTS ( TOP LEVEL )
261
262
263
264                          #       ###################
265                        #         #                 #
266                    #   #         #     CORRECT     #
267                     # #          #                 #
268                      #           ###################
269
270
271   Warning:  Ambiguity points were found and resolved arbitrarily.
272
273 LAYOUT CELL NAME:         HD
274 SOURCE CELL NAME:         HD
```

Checkpoints

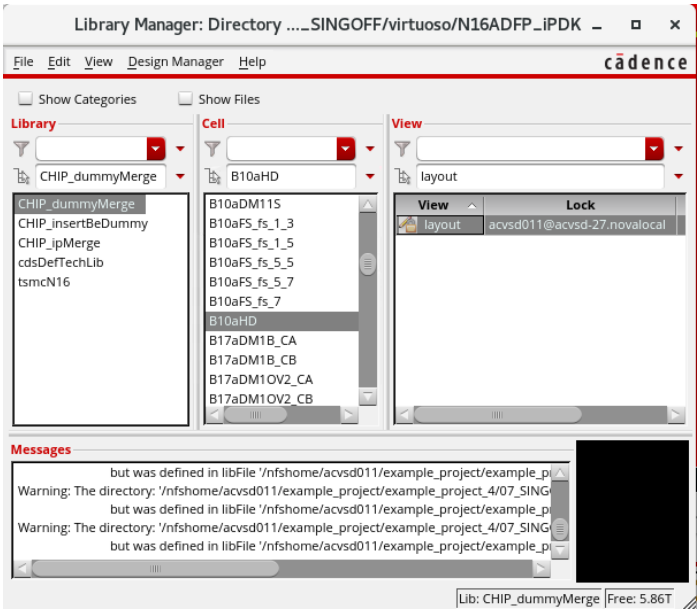
1. The setting of the PDK

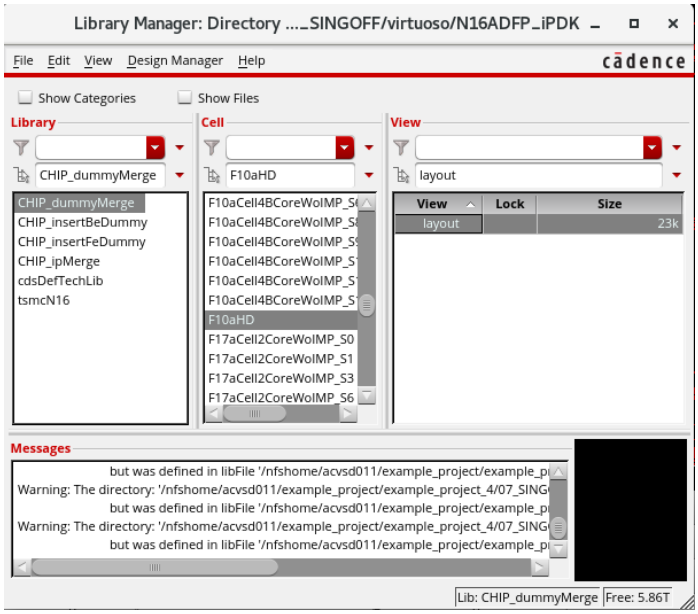
```
Please confirm with your selction information:
************************************************************
*** Select PDK type : Skill Pcell
*** Select process : LO
*** Select voltage : 0.8/1.8V
*** Select metal option : 1P11M_2Xa1Xd_h_3Xe_vhv_2Y2R UT-AlRDL
************************************************************
Are these correct (y|n) ?
y
```

2. The library is in Virtuoso after merging IP.

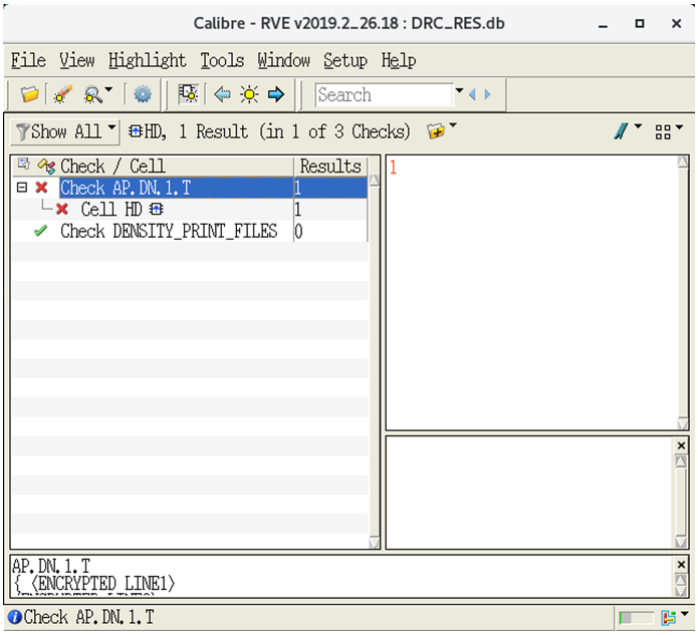


3. The layout view before dummy merging. Please zoom into the button-left corner and check the dummy metal.





4. The result of DRC check from the RVE browser in the Calibre embedded in Virtuoso



5. The result of the LVS check

```
259
260                  CELL COMPARISON RESULTS ( TOP LEVEL )
261
262
263
264                       #      ###################
265                     #        #                 #       *   *
266                   #   #      #     CORRECT     #         |
267                    # #       #                 #       \___/
268                     #        ###################
269
270
271   Warning:  Ambiguity points were found and resolved arbitrarily.
272
273 LAYOUT CELL NAME:         HD
274 SOURCE CELL NAME:         HD
275
```

6. **Due Tuesday, May. 26, 20:00**
7. Submit to NTU COOL in PDF format.